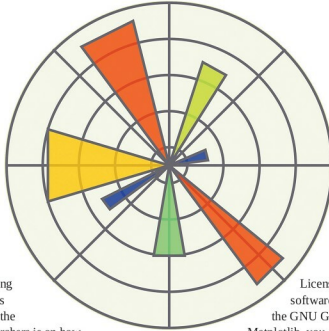


Scientific Graphics Visualisation: An Introduction to Matplotlib

Visualisation is the graphical display of information. Learning and cognition is enhanced by visualisation. While all of us are not born artists, scientific graphics visualisation merely requires the knowledge of code to visualise ideas and concepts.



The number of people doing research in our country is increasing daily. One of the challenges often faced by researchers is on how to best convey their findings to the research community and the general public. Well, there are no easy ways to express complicated scientific ideas in simple terms. The famous English idiom, 'A picture is worth a thousand words', gives us a hint. Images convey ideas in a simple but elegant manner. However, scientific visualisation is not limited to researchers alone—professionals working in diverse areas like IT, banking, automobiles and the healthcare sector also need good visualisation tools.

In this three-part series on scientific graphics visualisation, we will discuss three powerful tools used for this purpose. The tools are Matplotlib, PGF/TikZ and PSTricks. Though none of these tools can replace the other two completely, mastering even a single one of them will help researchers and professionals a lot. This series will help the reader skim through the basics of these three tools very quickly and then decide whether to master one, two or all three. All these tools are free and open source software. The additional advantage is that all three of them blend in perfectly with LaTeX, the best scientific document preparation tool.

Matplotlib

Matplotlib is a plotting library for the Python programming language licensed under the Python Software Foundation

License (PSF License), a free software license, compatible with the GNU General Public License. Using Matplotlib, you can generate plots, histograms, bar charts, scatter plots, etc, with the help of just a few lines of code. Matplotlib works in tandem with NumPy, a mathematical library for Python. It works as a multi-platform data visualisation library built on NumPy arrays and is designed to work within the Scipy stack. It is supported by Python 2, Python 3 and IPython. In fact, Matplotlib was first developed as a patch for IPython, for enabling interactive MATLAB-style plotting from the IPython shell.

Currently, I am using Fedora 24 and on searching, I found out that Python 3 is installed in my system, by default. But unlike Python 2 and IPython, the Matplotlib and NumPy packages were not part of my Python 3 installation; so I had to install them manually. Once the installation is successful, when the commands *python*, *python3* and *ipython* are executed in a terminal, you will be taken to the shells of Python 2, Python 3 and IPython, respectively. To use Matplotlib you need to import this package first with the command:

```
import matplotlib
```

The early development of Matplotlib was done by John D. Hunter and the first public version, Matplotlib version 0.1, was released in 2003. The latest stable release of Matplotlib is version 2.2.0, which was released on March 6, 2018. Figure 1